MATH1030 Assignment 4

复习笔记与详细解答

用途说明。这份文档按照 Assignment 4 对应的 lecture notes 范围整理，先给你一份中文复习笔记，再给出每题的详细解答。核心主题包括：**线性组合**、**线性相关/无关**、**向量空间与子空间**。

提交时必须注意。

- 正式提交只接受 **手写版**。
- **第 (1)–(13) 题**是课程评核必须提交的部分。
- 除非题目特别说明，否则作答时应写出适当的理由或计算步骤。
- 这份作业在论证中 **不可使用行列式**。
- **第 1、2 题**题目特别说明：提交版本不需要写 justification 或 calculation steps。

因此，这份 PDF 最适合作为 **复习讲义与参考答案**；真正交作业时，你可以再按课程要求整理成合适的手写版本。

依据你上传的 Assignment 4 与 lecture notes 范围整理。

2026 年 4 月 7 日

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1. 作业要求与核心主题

这份 Assignment 4 可以概括成四条主线：

1. **线性组合（linear combination）：**判断一个向量能否写成若干已知向量的组合。
2. **线性相关/无关（linear dependence / independence）：**判断是否存在非零系数使组合结果变成零向量。
3. **行化简、主元列与自由列：**用 row reduction 分析 span、线性相关与列之间的关系。
4. **子空间（subspace）：**用封闭性与零向量判断一个集合是不是向量空间中的子空间。

整份作业其实可以用下面这三组对应关系来统一理解：

$$\text{线性组合问题} \iff \text{求解}[u_1 \ \cdots \ u_k]c = z$$

$$\text{线性相关问题} \iff \text{求解}[u_1 \ \cdots \ u_k]c = 0 \text{ 是否有非零解}$$

$$\text{子空间问题} \iff \text{检验}0 \in S,\ x + y \in S,\ \alpha x \in S.$$

2. 先看复习笔记

2.1. 线性组合与矩阵观点

设 $u_1, \ldots, u_k$ 是同型列向量。若存在标量 $c_1, \ldots, c_k$ 使得

$$z = c_1 u_1 + \cdots + c_k u_k,$$

则称 z 是 $u_1, \ldots, u_k$ 的一个线性组合。

如果把这些向量排成一个矩阵

$$U = [\, u_1 \ u_2 \ \cdots \ u_k \,], \qquad c = \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix},$$

那么上面的式子就等价于

$$Uc = z.$$

所以：**凡是“是否为线性组合”的题目，本质上都可以转成一个线性方程组来解。**

2.2. 线性相关/无关与齐次系统

向量组 $u_1, \ldots, u_k$ 线性相关，意思是存在一组**不全为零**的系数

$$c_1, \ldots, c_k$$

使得

$$c_1 u_1 + \cdots + c_k u_k = 0.$$

矩阵写法就是

$$\begin{bmatrix} u_1 & \cdots & u_k \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix} = 0.$$

因此：

- 若对应齐次系统有**非零解**，则这些向量**线性相关**；
- 若对应齐次系统只有**零解**，则这些向量**线性无关**。

这正是第 3、4、6、9 题反复出现的核心思想。

2.3. 行化简、主元列与自由列怎么用

对矩阵 $U = [u_1 \ \cdots \ u_k]$ 做行化简时，要特别注意：

- **主元列（pivot column）**对应 leading variable；
- 如果出现 **自由列（free column）**，那么齐次系统 $Uc = 0$ 一定有非零解，因此列向量一定线性相关；
- 在 RREF 里，**非主元列可以写成主元列的线性组合**，而且系数通常可以直接从该列读出来。

所以：

- 判断线性相关，关键看有没有 free column；
- 求“某列如何由其他列表示”，关键看 RREF 中该列相对于 pivot columns 的表达。

第 6 题和第 9 题就是这套方法的典型应用。

2.4. 矩阵会保持线性组合

如果已经知道

$$v = c_1u_1 + \cdots + c_ku_k,$$

那么对任意适当大小的矩阵 B，都有

$$Bv = c_1Bu_1 + \cdots + c_kBu_k.$$

也就是说，**线性变换会把“原来的线性组合关系”完整保留下来**。

这条性质在第 1(b)、第 8(b) 里都直接用到。

反过来，有一个很重要的“可逆/单射”思路：

> 如果 $Cw = 0$ 只可能推出 $w = 0$，那么 C 没有非零核向量，也就是对向量是单射的。因此一旦
>
> $$Cz = C(\text{某个线性组合}),$$
>
> 就可以推出
>
> $$z = \text{同一个线性组合}.$$

这正是第 8(c) 的逻辑基础。

2.5. 子空间判定法

设 S 是某个向量空间的子集。要证明 S 是子空间，最稳妥的方法就是检查三件事：

1. $0 \in S$；
2. 若 $x, y \in S$，则 $x + y \in S$；
3. 若 $x \in S$ 且 $\alpha \in \mathbb{R}$，则 $\alpha x \in S$。

常见结论要记住：

- 零空间、解空间、null space 通常都是子空间；
- 用线性方程“等于零”刻画出来的集合，往往是子空间；
- 两个子空间的**和**通常还是子空间；
- 两个子空间的**并**一般**不是**子空间，除非其中一个包含另一个；
- 用不等式定义的集合通常**不是**子空间，因为乘以负数时很容易失去封闭性。

第 10–13 题几乎都可以用这些模板快速判断。

2.6. 做这份作业时最值得记住的几点

1. 第 1、2 题提交时只需写答案，但你复习时还是要知道背后的理由。
2. 第 3–9 题的关键是：**把问题改写成矩阵方程，再用 row reduction**。
3. 第 10–13 题不要一上来就硬算，优先用**子空间定义**与**闭合性**判断。
4. 因为这份作业不允许在论证中用 determinant，所以请尽量使用 **pivot / free column / homogeneous system / null space** 这些语言。
5. 当题目要求“with direct reference to the definition”时，写作时一定要明确点出你所用的定义，而不只是给结论。

3. Full solutions

Question 1

Given

$$A = [\,a_1\ a_2\ a_3\ a_4\ a_5\,], \qquad u = \begin{bmatrix} 4 \\ 8 \\ -5 \\ 3 \\ 2 \end{bmatrix}, \qquad v = Au.$$

(a) Since $v = Au$, by column interpretation of matrix multiplication,

$$\boxed{v = 4a_1 + 8a_2 - 5a_3 + 3a_4 + 2a_5}.$$

(b) Yes. Applying A^9 to the relation in part (a),

$$A^9 v = 4A^9 a_1 + 8A^9 a_2 - 5A^9 a_3 + 3A^9 a_4 + 2A^9 a_5.$$

Hence

$$\boxed{A^9 v \text{ is a linear combination of } A^9 a_1, \ldots, A^9 a_5}.$$

(c) If $y = -3x$, then

$$a_4 = \begin{bmatrix} -6 \\ -9 \\ 0 \\ y \\ 3 \end{bmatrix} = \begin{bmatrix} -6 \\ -9 \\ 0 \\ -3x \\ 3 \end{bmatrix} = -3 \begin{bmatrix} 2 \\ 3 \\ 0 \\ x \\ -1 \end{bmatrix} = -3a_2.$$

So

$$v = 4a_1 + 8a_2 - 5a_3 + 3(-3a_2) + 2a_5 = 4a_1 - a_2 - 5a_3 + 2a_5.$$

Therefore

$$\boxed{v = 4a_1 - a_2 - 5a_3 + 2a_5}.$$

Question 2

The system is

$$\begin{cases} 3x_1 + 5x_4 + x_5 = 0, \\ x_1 - 2x_2 + 7x_4 + x_5 = 0, \\ -x_1 + x_2 + 3x_4 + 3x_5 = 0, \\ -x_1 + x_2 + x_4 + 2x_5 = 0, \\ x_2 + x_3 + x_4 = 0. \end{cases}$$

(a) Its coefficient matrix is

$$\boxed{\begin{bmatrix} 3 & 0 & 0 & 5 & 1 \\ 1 & -2 & 0 & 7 & 1 \\ -1 & 1 & 0 & 3 & 3 \\ -1 & 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}}$$

(b) Let

$$t = \begin{bmatrix} 1 \\ -2 \\ a \\ -1 \\ b \end{bmatrix}.$$

Substitute into the system:

$$3(1) + 5(-1) + b = 0 \implies b = 2.$$

Then from

$$-(-2) + a + (-1) = 0$$

we get

$$a = 3.$$

Hence

$$\boxed{a = 3, \qquad b = 2.}$$

(c)(i) Note that the coefficient matrix in part (a) is exactly

$$[\, u_1 \ u_2 \ u_3 \ u_4 \ u_5 \,].$$

From part (b), the vector

$$\begin{bmatrix} 1 \\ -2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$$

is a non-trivial solution of the corresponding homogeneous system. Therefore

$$u_1 - 2u_2 + 3u_3 - u_4 + 2u_5 = 0.$$

So the five vectors are linearly dependent.

One non-trivial linear relation is

$$\boxed{u_1 - 2u_2 + 3u_3 - u_4 + 2u_5 = 0}.$$

(c)(ii) Rearranging,

$$2u_5 = -u_1 + 2u_2 - 3u_3 + u_4,$$

so

$$\boxed{u_5 = -\tfrac{1}{2}u_1 + u_2 - \tfrac{3}{2}u_3 + \tfrac{1}{2}u_4}.$$

Hence u_5 is a linear combination of u_1, u_2, u_3, u_4.

Question 3

Let

$$u_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \\ -2 \end{bmatrix}, \quad u_2 = \begin{bmatrix} -3 \\ 0 \\ -1 \\ -1 \\ 2 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \\ -3 \end{bmatrix}, \quad z = \begin{bmatrix} 5 \\ 7 \\ 4 \\ 0 \\ -1 \end{bmatrix}.$$

(a) Solve

$$c_1u_1 + c_2u_2 + c_3u_3 = z.$$

A solution is

$$c_1 = -3, \qquad c_2 = -2, \qquad c_3 = 1.$$

Indeed,

$$-3u_1 - 2u_2 + u_3 = \begin{bmatrix} -3 \\ 6 \\ 0 \\ -3 \\ 6 \end{bmatrix} + \begin{bmatrix} 6 \\ 0 \\ 2 \\ 2 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 4 \\ 0 \\ -1 \end{bmatrix} = z.$$

Therefore

$$\boxed{z = -3u_1 - 2u_2 + u_3}.$$

(b) From part (a),

$$-2u_2 = z + 3u_1 - u_3.$$

Hence

$$u_2 = -\tfrac{1}{2}z - \tfrac{3}{2}u_1 + \tfrac{1}{2}u_3.$$

By direct reference to the definition of linear combination, this shows that u_2 is a linear combination of u_1, u_3, z.

So the answer is **yes**, with

$$\boxed{u_2 = -\tfrac{3}{2}u_1 + \tfrac{1}{2}u_3 - \tfrac{1}{2}z}.$$

Question 4

Let

$$v_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \\ 3 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 3 \\ -2 \\ 1 \\ 2 \\ -1 \end{bmatrix}.$$

(a) A non-trivial relation is

$$-6v_1 + 9v_2 - 7v_3 + v_4 = 0.$$

Since the coefficients are not all zero, the vectors are linearly dependent.

Hence

$$\boxed{v_1, v_2, v_3, v_4 \text{ are linearly dependent}}.$$

One acceptable linear relation is

$$\boxed{-6v_1 + 9v_2 - 7v_3 + v_4 = 0}.$$

(b) We have

$$w_1 = -v_1 + v_2, \qquad w_2 = v_2 - 2v_3, \qquad w_3 = v_3 - v_4.$$

Now compute

$$6w_1 + 3w_2 - w_3 = 6(-v_1 + v_2) + 3(v_2 - 2v_3) - (v_3 - v_4) = -6v_1 + 9v_2 - 7v_3 + v_4 = 0.$$

Since not all coefficients $6, 3, -1$ are zero, the vectors w_1, w_2, w_3 are linearly dependent by definition.

Thus

$$\boxed{w_1, w_2, w_3 \text{ are linearly dependent}}.$$

A non-trivial relation is

$$\boxed{6w_1 + 3w_2 - w_3 = 0}.$$

Question 5

Suppose

$$v = \begin{bmatrix} 7 \\ 3 \\ \alpha \\ 6 \end{bmatrix}, \qquad u = \begin{bmatrix} 2 \\ 0 \\ 1 \\ \beta \end{bmatrix}, \qquad t = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix},$$

and suppose v is a linear combination of u and t:

$$v = cu + dt.$$

Comparing the second entry gives

$$d = 3.$$

Comparing the first entry gives

$$2c + 2d = 7 \implies 2c + 6 = 7 \implies c = \frac{1}{2}.$$

Then comparing the third entry gives

$$\alpha = c = \frac{1}{2}.$$

Finally, comparing the fourth entry gives

$$6 = c\beta + d = \frac{1}{2}\beta + 3,$$

so

$$\beta = 6.$$

Therefore

$$\boxed{\alpha = \frac{1}{2}, \qquad \beta = 6.}$$

Question 6

Let

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ \alpha \\ 0 \\ \beta \end{bmatrix}, \quad u_2 = \begin{bmatrix} 1 \\ 1 \\ \alpha \\ 1 \\ \beta \end{bmatrix}, \quad u_3 = \begin{bmatrix} -1 \\ 1 \\ -3 \\ 2 \\ -2 \end{bmatrix}, \quad u_4 = \begin{bmatrix} 3 \\ 5 \\ -\alpha \\ 7 \\ \beta^2 \end{bmatrix},$$

and

$$U = [\, u_1 \; u_2 \; u_3 \; u_4 \,].$$

(a) We start with

$$U = \begin{bmatrix} 1 & 1 & -1 & 3 \\ 0 & 1 & 1 & 5 \\ \alpha & \alpha & -3 & -\alpha \\ 0 & 1 & 2 & 7 \\ \beta & \beta & -2 & \beta^2 \end{bmatrix}.$$

Apply row operations:

$$R_3 \leftarrow R_3 - \alpha R_1, \qquad R_4 \leftarrow R_4 - R_2, \qquad R_5 \leftarrow R_5 - \beta R_1.$$

This gives

$$\begin{bmatrix} 1 & 1 & -1 & 3 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & \alpha-3 & -4\alpha \\ 0 & 0 & 1 & 2 \\ 0 & 0 & \beta-2 & \beta^2-3\beta \end{bmatrix}.$$

Now apply

$$R_3 \leftarrow R_3 - (\alpha-3)R_4, \qquad R_5 \leftarrow R_5 - (\beta-2)R_4,$$

to get

$$\begin{bmatrix} 1 & 1 & -1 & 3 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & 6-6\alpha \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & \beta^2-5\beta+4 \end{bmatrix}.$$

Swap R_3 and R_4, then multiply the new R_4 by $-\frac{1}{6}$. Hence U is row-equivalent to

$$U^{\#} = \begin{bmatrix} 1 & 1 & -1 & 3 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & \alpha-1 \\ 0 & 0 & 0 & \beta^2-5\beta+4 \end{bmatrix}.$$

Therefore

$$\boxed{a=-1, \qquad b=-5, \qquad c=4.}$$

(b) The first three columns are pivot columns already. So there is a free column exactly when column 4 is *not* a pivot column, i.e. when both entries beneath it vanish:

$$\alpha-1=0, \qquad \beta^2-5\beta+4=0.$$

Thus

$$\beta^2-5\beta+4=(\beta-1)(\beta-4),$$

so the required values are

$$\boxed{(\alpha,\beta)=(1,1) \text{ or } (1,4).}$$

(c) The vectors u_1, u_2, u_3, u_4 are linearly dependent exactly when U has a free column, so exactly for

$$\boxed{(\alpha,\beta)=(1,1) \text{ or } (1,4).}$$

In either case, solving the homogeneous system gives the non-trivial relation

$$-2u_1-3u_2-2u_3+u_4=0.$$

Equivalently,

$$\boxed{u_4=2u_1+3u_2+2u_3}.$$

Question 7

We must prove:

> Suppose u_1, u_2, u_3, u_4, u_5 are column vectors with p entries. Then every scalar multiple of every linear combination of u_1, u_2, u_3, u_4, u_5 is a linear combination of u_1, u_2, u_3, u_4, u_5.

Proof. Let

$$w$$

be a linear combination of u_1, u_2, u_3, u_4, u_5. By definition, there exist scalars $c_1, \ldots, c_5$ such that

$$w = c_1u_1 + c_2u_2 + c_3u_3 + c_4u_4 + c_5u_5.$$

Let $\lambda \in \mathbb{R}$. Then

$$\lambda w = \lambda(c_1u_1 + \cdots + c_5u_5) = (\lambda c_1)u_1 + \cdots + (\lambda c_5)u_5.$$

Since $\lambda c_1, \ldots, \lambda c_5$ are again scalars, the vector λw is a linear combination of u_1, u_2, u_3, u_4, u_5.

Hence the statement is proved. □

Question 8

Let

$$v = \begin{bmatrix} 3 \\ 3 \\ 2 \\ 2 \\ 5 \\ 5 \end{bmatrix}, \quad u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad u_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

(a) One expression is

$$\boxed{v = 2u_1 + 2u_2 + u_3 + 3u_4}.$$

(b) Yes. Since matrix multiplication preserves linear combinations,

$$Bv = B(2u_1 + 2u_2 + u_3 + 3u_4) = 2Bu_1 + 2Bu_2 + Bu_3 + 3Bu_4.$$

So

$$\boxed{Bv \text{ is a linear combination of } Bu_1, Bu_2, Bu_3, Bu_4.}$$

(c) Since $\mathrm{LS}(C, 0_6)$ has no non-trivial solution, the equation

$$Cw = 0$$

implies

$$w = 0.$$

From part (a),

$$v = 2u_1 + 2u_2 + u_3 + 3u_4.$$

Using the assumptions $Cv' = v$ and $Cu'_j = u_j$, we get

$$C\big(v' - (2u'_1 + 2u'_2 + u'_3 + 3u'_4)\big) = Cv' - (2Cu'_1 + 2Cu'_2 + Cu'_3 + 3Cu'_4) = v - (2u_1 + 2u_2 + u_3 + 3u_4) = 0.$$

By triviality of the null space of C,

$$v' - (2u'_1 + 2u'_2 + u'_3 + 3u'_4) = 0.$$

Hence

$$\boxed{v' = 2u'_1 + 2u'_2 + u'_3 + 3u'_4}.$$

So the answer is **yes**.

Question 9

The reduced row-echelon form is

$$A' = \begin{bmatrix} 1 & 0 & -3 & 0 & 2 & 0 & 0 & -2 \\ 0 & 1 & 4 & 0 & -5 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 7 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 6 \end{bmatrix}.$$

(a) The pivot columns of A' are

$$\boxed{1,\ 2,\ 4,\ 6,\ 7.}$$

(b)(i) Reading non-pivot columns from A' in terms of pivot columns:

$$\boxed{a_3 = -3a_1 + 4a_2}$$

$$\boxed{a_5 = 2a_1 - 5a_2 + 7a_4}$$

$$\boxed{a_8 = -2a_1 - 3a_2 - 3a_4 - 3a_6 + 6a_7}.$$

(b)(ii) Since

$$a_3 = -3a_1 + 4a_2,$$

we get

$$a_2 = \frac{1}{4}(a_3 + 3a_1) = \frac{1}{4}\begin{bmatrix} -3 \\ -1 \\ -5 \\ 0 \\ 5 \end{bmatrix} + \frac{3}{4}\begin{bmatrix} 1 \\ -1 \\ 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 2 \end{bmatrix}.$$

So

$$\boxed{a_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 2 \end{bmatrix}}.$$

Then

$$a_5 = 2a_1 - 5a_2 + 7a_4 = 2\begin{bmatrix} 1 \\ -1 \\ 3 \\ 0 \\ 1 \end{bmatrix} - 5\begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} + 7\begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ -11 \\ 1 \\ 21 \\ 20 \end{bmatrix}.$$

Hence

$$\boxed{a_5 = \begin{bmatrix} 9 \\ -11 \\ 1 \\ 21 \\ 20 \end{bmatrix}}.$$

Finally,

$$a_8 = -2a_1 - 3a_2 - 3a_4 - 3a_6 + 6a_7$$

gives

$$a_8 = -2\begin{bmatrix} 1 \\ -1 \\ 3 \\ 0 \\ 1 \end{bmatrix} - 3\begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} - 3\begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \\ 4 \end{bmatrix} - 3\begin{bmatrix} 0 \\ 9 \\ 11 \\ 5 \\ 8 \end{bmatrix} + 6\begin{bmatrix} 7 \\ -3 \\ 4 \\ 0 \\ 15 \end{bmatrix} = \begin{bmatrix} 37 \\ -34 \\ -18 \\ -24 \\ 46 \end{bmatrix}.$$

Thus

$$\boxed{a_8 = \begin{bmatrix} 37 \\ -34 \\ -18 \\ -24 \\ 46 \end{bmatrix}}.$$

(c) Since the pivot columns of A' are e_1, e_2, e_3, e_4, e_5, the matrix G must have columns

$$a_1,\ a_2,\ a_4,\ a_6,\ a_7.$$

Therefore one suitable invertible matrix is

$$\boxed{G = \begin{bmatrix} 1 & 0 & 1 & 0 & 7 \\ -1 & -1 & -2 & 9 & -3 \\ 3 & 1 & 0 & 11 & 4 \\ 0 & 0 & 3 & 5 & 0 \\ 1 & 2 & 4 & 8 & 15 \end{bmatrix}}$$

because

$$A = GA'.$$

Question 10

Define

$$V = \left\{ x \in \mathbb{R}^7 : \text{ there exist } t \in W,\ y \in \mathbb{R}^7 \text{ such that } x = At + y \text{ and } By = 0_5 \right\}.$$

(a) The statement is true.

Take any $x \in V$ and any $\alpha \in \mathbb{R}$. Then there exist $t \in W$ and $y \in \mathbb{R}^7$ such that

$$x = At + y, \qquad By = 0_5.$$

Since W is a subspace, $\alpha t \in W$. Also

$$B(\alpha y) = \alpha(By) = \alpha 0_5 = 0_5.$$

And

$$\alpha x = \alpha(At + y) = A(\alpha t) + \alpha y.$$

Therefore $\alpha x \in V$.

So

$$\boxed{\text{Yes, the statement in (a) is true.}}$$

(b) Yes, V is a subspace of $\mathbb{R}^7$.

First, $0 \in V$: choose $t = 0_9 \in W$ and $y = 0_7$. Then

$$0 = A0 + y, \qquad By = 0.$$

Now let $x_1, x_2 \in V$. Then there exist $t_1, t_2 \in W$ and $y_1, y_2 \in \mathbb{R}^7$ such that

$$x_1 = At_1 + y_1, \quad By_1 = 0, \qquad x_2 = At_2 + y_2, \quad By_2 = 0.$$

Since W is a subspace, $t_1 + t_2 \in W$. Also

$$B(y_1 + y_2) = By_1 + By_2 = 0 + 0 = 0.$$

Thus

$$x_1 + x_2 = A(t_1 + t_2) + (y_1 + y_2) \in V.$$

Part (a) already gives closure under scalar multiplication. Hence V is a subspace.

So

$$\boxed{V \text{ is a subspace of } \mathbb{R}^7.}$$

Question 11

Let

$$W = \{X \in M_{6,7}(\mathbb{R}) : AXB \in V\},$$

where V is a subspace of $M_{5,8}(\mathbb{R})$.

(a) Since

$$AO_{6\times 7}B = O_{5\times 8},$$

and V is a subspace, it contains the zero matrix. Hence

$$O_{6\times 7} \in W.$$

(b) Suppose $X \in W$ and $\kappa \in \mathbb{R}$. Then $AXB \in V$. Since V is closed under scalar multiplication,

$$\kappa(AXB) \in V.$$

But

$$\kappa(AXB) = A(\kappa X)B.$$

Hence $\kappa X \in W$.

(c) Yes, W is a subspace of $M_{6,7}(\mathbb{R})$.

We already know $O_{6\times 7} \in W$. Now let $X, Y \in W$. Then $AXB \in V$ and $AYB \in V$. Since V is a subspace,

$$AXB + AYB \in V.$$

But

$$AXB + AYB = A(X + Y)B.$$

Hence $X + Y \in W$.

Part (b) gives closure under scalar multiplication. Therefore W is a subspace.

(d) Yes. Since $M_{6,7}(\mathbb{R})$ is a vector space over $\mathbb{R}$, and W is a subspace of it, W is itself a vector space over $\mathbb{R}$.

Therefore

$$\boxed{W \text{ is a subspace of } M_{6,7}(\mathbb{R}), \text{ hence also a vector space over } \mathbb{R}.}$$

Question 12

Define

$$T = \left\{ y \in \mathbb{R}^3 : \text{ the sum of the entries of } y \text{ is non-negative} \right\}.$$

(a) T is *not* a subspace of $\mathbb{R}^3$.

For example,

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \in T$$

because the sum of its entries is $1 \geq 0$. But

$$-\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \notin T$$

because the sum of its entries is $-1 < 0$.

So T is not closed under scalar multiplication. Hence

$$\boxed{T \text{ is not a subspace of } \mathbb{R}^3.}$$

(b) The set

$$S = \left\{ x \in \mathbb{R}^5 : \text{ there exists } b \in T \text{ such that } x \text{ solves } \mathrm{LS}(A, b) \right\}$$

is *not* a subspace of $\mathbb{R}^5$.

Because $\mathrm{LS}(A, b)$ is consistent for every $b \in \mathbb{R}^3$, we can choose $x \in \mathbb{R}^5$ such that

$$Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Since $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \in T$, this gives $x \in S$.

But then

$$A(-x) = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix},$$

and $\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \notin T$. So $-x \notin S$.

Thus S is not closed under scalar multiplication, hence not a subspace. Therefore

$$\boxed{S \text{ is not a subspace of } \mathbb{R}^5.}$$

Question 13

Let

$$A = [\,1\ p\ q\ r\,], \qquad B = [\,1\ s\ t\ u\,], \qquad p \neq s,$$

and

$$K = \mathrm{N}(A) \cup \mathrm{N}(B) \subseteq \mathbb{R}^4.$$

(a) Take

$$x = \begin{bmatrix} -p \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Then

$$Ax = (-p) + p = 0,$$

so $x \in \mathrm{N}(A)$. Hence $x \in K$.

But

$$Bx = (-p) + s = s - p \neq 0$$

because $p \neq s$. So $Bx \neq 0$.

Therefore:

$$\boxed{x = \begin{bmatrix} -p \\ 1 \\ 0 \\ 0 \end{bmatrix} \in \mathrm{N}(A) \subseteq K, \qquad Bx = s - p \neq 0.}$$

(b) Take

$$y = \begin{bmatrix} -s \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Then

$$By = (-s) + s = 0,$$

so $y \in \mathrm{N}(B)$. Hence $y \in K$.

But

$$Ay = (-s) + p = p - s \neq 0.$$

Therefore:

$$\boxed{y = \begin{bmatrix} -s \\ 1 \\ 0 \\ 0 \end{bmatrix} \in \mathrm{N}(B) \subseteq K, \qquad Ay = p - s \neq 0.}$$

(c) Consider the linear combination

$$x - y = \begin{bmatrix} -p \\ 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -s \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} s-p \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since $p \neq s$, this vector is non-zero. Also

$$A(x-y) = s - p \neq 0, \qquad B(x-y) = s - p \neq 0.$$

So

$$x - y \notin \mathrm{N}(A) \quad \text{and} \quad x - y \notin \mathrm{N}(B).$$

Hence

$$x - y \notin K.$$

Thus one appropriate example is

$$\boxed{x - y \notin K.}$$

(d) Since $x, y \in K$ but $x - y \notin K$, the set K is not closed under addition. Therefore

$$\boxed{K \text{ is not a subspace of } \mathbb{R}^4.}$$

4. Final quick-answer list

For fast handwritten rewriting, here is a compressed list of the final conclusions.

1. $v = 4a_1 + 8a_2 - 5a_3 + 3a_4 + 2a_5$; $\quad A^9 v = 4A^9 a_1 + 8A^9 a_2 - 5A^9 a_3 + 3A^9 a_4 + 2A^9 a_5$; if $y = -3x$, then $v = 4a_1 - a_2 - 5a_3 + 2a_5$.

2. Coefficient matrix: $\begin{bmatrix} 3 & 0 & 0 & 5 & 1 \\ 1 & -2 & 0 & 7 & 1 \\ -1 & 1 & 0 & 3 & 3 \\ -1 & 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$; $\quad a = 3, b = 2$; $\quad u_1 - 2u_2 + 3u_3 - u_4 + 2u_5 = 0$; $\quad u_5 = -\frac{1}{2}u_1 + u_2 - \frac{3}{2}u_3 + \frac{1}{2}u_4$.

3. $z = -3u_1 - 2u_2 + u_3$; $\quad$ yes, $u_2 = -\frac{3}{2}u_1 + \frac{1}{2}u_3 - \frac{1}{2}z$.

4. Dependent; one relation: $-6v_1 + 9v_2 - 7v_3 + v_4 = 0$. Also w_1, w_2, w_3 dependent with $6w_1 + 3w_2 - w_3 = 0$.

5. $\alpha = \frac{1}{2}$, $\beta = 6$.

6. $a = -1$, $b = -5$, $c = 4$; free column iff $(\alpha, \beta) = (1, 1)$ or $(1, 4)$; in either case $-2u_1 - 3u_2 - 2u_3 + u_4 = 0$.

7. Scalar multiple of a linear combination is again a linear combination.

8. $v = 2u_1 + 2u_2 + u_3 + 3u_4$; yes for Bv; yes for v', with $v' = 2u'_1 + 2u'_2 + u'_3 + 3u'_4$.

9. Pivot columns: $1, 2, 4, 6, 7$; $a_3 = -3a_1 + 4a_2$, $a_5 = 2a_1 - 5a_2 + 7a_4$, $a_8 = -2a_1 - 3a_2 - 3a_4 - 3a_6 + 6a_7$; $a_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 2 \end{bmatrix}$, $a_5 = \begin{bmatrix} 9 \\ -11 \\ 1 \\ 21 \\ 20 \end{bmatrix}$, $a_8 = \begin{bmatrix} 37 \\ -34 \\ -18 \\ -24 \\ 46 \end{bmatrix}$; $G = [\, a_1 \ a_2 \ a_4 \ a_6 \ a_7 \,]$.

10. Part (a): true. $\quad$ Part (b): V is a subspace of $\mathbb{R}^7$.

11. $O_{6\times 7} \in W$; scalar closure true; W is a subspace of $M_{6,7}(\mathbb{R})$ and hence a vector space.

12. T is not a subspace; S is not a subspace.

13. $x = \begin{bmatrix} -p \\ 1 \\ 0 \\ 0 \end{bmatrix} \in \mathrm{N}(A) \subseteq K$, $Bx \neq 0$; $\quad y = \begin{bmatrix} -s \\ 1 \\ 0 \\ 0 \end{bmatrix} \in \mathrm{N}(B) \subseteq K$, $Ay \neq 0$; $\quad x - y \notin K$; hence K is not a subspace.